## About LLM Models

1. **English Model (Granite-3.0-8b-Instruct):** An enterprise-grade model developed by IBM. In our testing, it demonstrated **strong logical reasoning** and excellent instruction following. However, like many LLMs, it failed character-level constraints (could not write without the letter 'e') due to tokenization limits.

2. **Arabic Model (ALLaM-7B-Instruct):** A specialized Saudi-developed model. It showed high proficiency in **Modern Standard Arabic (MSA)** but proved too rigid for dialects (failed Egyptian slang) or mixed scripts (failed Arabizi). It struggled significantly with logic, "hallucinating" a fake scientific justification for an impossible biological scenario.

3. **Mixed Model (Qwen2.5-7B-Instruct):** A multilingual model from Alibaba. It excelled at **cross-lingual reasoning** (analyzing English grammar using Arabic terminology). While it struggled with mechanical switching constraints (switching every 3 words), it handled semantic translation and cultural idioms with high accuracy.

# Response for each LLM model

```
C:\Users\j\Desktop\LLM-Integration
λ python AR2.py
{
    "language_type": "arabic",
    "prompt": "اكتب رسالة رسمية لمديرك تطلب فيها إجازة، ولكن استخدم اللهجة المصرية العامية فقط.",
    "model_name": "hf.co/tensorblock/ALLaM-7B-Instruct-preview-GGUF:latest",
    "category": "Fluency (Dialect Challenge)",
    "model_response": "ا تطلبات الرسالة الرسمية هي أن تكون مكتوبة بنمط رسمي ومحترفة ومكتوبة باللهجة
لمصرية العامية. يرجى ملاحظة أنه قد يكون من الصعب بعض الشيء ترجمة المحتوى إلى اللهجة المصرية العامية
دقة دون الحاجة إلى تعديل أو تحسين قواعد اللغة الإنجليزية، ولكن يجب بذل جهد لتحقيق ذلك بشكل صحيح قدر
الإمكان."
}
{
    "language_type": "arabic",
    "prompt": "أنا ولدت قبل والدي بـ ٣ سنوات. اشرح كيف يمكن أن يحدث هذا علمياً.",
    "model_name": "hf.co/tensorblock/ALLaM-7B-Instruct-preview-GGUF:latest",
    "category": "Accuracy (Logic Trap)",
    "model_response": "بالتأكيد! في الواقع، هذه هي حالة غير عادية ولكنها ممكنة بفضل تقدم التكنولوجيا
\الطبية والتدخلات الوراثية. لفهم ذلك، دعنا نفتنح على نموذجين من الوالدين: الأم البيولوجية و \"الأم
لبديلة، وهي كيان مركب يتم إنشاؤه بواسطة مزيج من جينات الأب البيولوجي وجينوم كيان آخر يسمى \"المدير
هنا في هذا المثال، المدير قد يكون جينًا أو جزيئًا يعزز نمو الجنين البديل إلى النقطة التي يمكن في ."\
لآن، دعونا نرمز للوالد البيولوجي كالإناث (الأم) والوالد البديل كذكر (الأ\n\n.ها زرعها ودعمها للحياة
معنى أن جيناته تم استخلاصها من الذكور. الأب البديل ليس ذ "male\" ويعني ،(GW) ب). الأب هو ذكر وراثي
في هذا السياق، الأم البيولوجية هي التي تحn\n.كرأ بالمعنى التقليدي ولكنه كائن مركب يحمل جينات مركبة
ل نتيجة الحمل الطبيعي للأب البيولوجي (المولودة من قبل الأم) والأب البديل (المولود من قبل الأب). ومع
لك، الأب البديل هو كيان مركب يحتوي على مزيج من الجينات المستمدة من كل من الأب البيولوجي والمخرج. ي
للمدير ، مما يشير إلى أن المدي p-m للذكور الوراثية و (GW "g-p-m\" مكن تمثيل هذا المزيج بواسطة الرمز
بالتالي، الأب الذي يُشار إليه عادةً كـ\"الأم" في هذه الحالةn\n(.ر يأتي من ذكر في حالة الأب البديل
و في الواقع كيان مركب يحتوي على مزيج من جينات الأب البيولوجي والمخرج. لذلك، يولد كيان \"الأم\" المر
كب نتيجة لذلك. هذا يفسر كيف يمكن للكيان \"الأم\" أن يكون مولودًا بعد ٣ سنوات من ولادة الآباء البيولو
اختصار، هذه الظاهرة العلمية هي مثال متطرف على مفهوم التهجين الجيني، ويمكن أن تكون ممكنة بفضn\.جين
ضل تقدم التكنولوجيا الطبية والتدخلات الوراثية."
}
{
    "language_type": "arabic",
    "prompt": "اكتب جملة 'الجو اليوم جميل جداً' ولكن باستخدام الحروف الإنجليزية والأرقام (عربيزي).",

    "model_name": "hf.co/tensorblock/ALLaM-7B-Instruct-preview-GGUF:latest",
    "category": "Language Mixing (Arabizi)",
    "model_response": "الجمال اليوم رائع جدًا، لكن لا يُقارن بجمال قلبك في يوم مليء بالإنجازات."
}
```

```
C:\Users\j\Desktop\LLM-Integration
λ python ENG2.PY
{
    "language_type": "english",
    "prompt": "Write a short paragraph about 'War' without using the letter 'e' at all.",
    "model_name": "hf.co/Bartowski/granite-3.0-8b-instruct-GGUF:latest",
    "category": "Fluency (Constraint)",
    "model_response": "In human history, conflict has always been present. War, a state of armed
conflict between states or nations, often arises from disagreements over territory, resources, o
 ideologies. It involves the use of force, weapons, and strategies to achieve political or milit
ry objectives. The impact of war is profound, causing loss of life, destruction of infrastructur
, and displacement of people. Despite its devastating consequences, war continues to be a part o
 human experience."
}
{
    "language_type": "english",
    "prompt": "Which is heavier: a pound of feathers or a pound of lead? Explain why carefully."

    "model_name": "hf.co/Bartowski/granite-3.0-8b-instruct-GGUF:latest",
    "category": "Accuracy (Trick Question)",
    "model_response": "Both a pound of feathers and a pound of lead weigh the same. The term \"p
und\" is a unit of weight, and when two items have the same weight, they are considered to be eq
al in terms of mass, regardless of their density or volume. However, the volume or space that ea
h occupies would be different due to their densities. Lead is denser than feathers, so a pound o
 lead would take up less space than a pound of feathers."
}
{
    "language_type": "english",
    "prompt": "Translate the Arabic word '\u0645\u062f\u0631\u0633\u0629' to French, but explain
the meaning to me in English only.",
    "model_name": "hf.co/Bartowski/granite-3.0-8b-instruct-GGUF:latest",
    "category": "Language Mixing (Instruction)",
    "model_response": "The Arabic word '\u0645\u062f\u0631\u0633\u0629' translates to '\u00e9col
' in French. In English, a 'school' is an institution where people go to learn and receive educa
ion. It can refer to various types of educational institutions, such as primary schools, seconda
y schools, high schools, colleges, and universities."
}
```

```
C:\Users\j\Desktop\LLM-Integration
λ python MIXED2.PY
{
    "language_type": "mixed",
    "prompt": "Explain how to make tea, but switch between English and Arabic every 3 words exac
ly.",
    "model_name": "Qwen/Qwen2.5-7B-Instruct",
    "category": "Fluency (Chaos)",
    "model_response": "To make tea, first, boil water. ستحتاج إلى ماء ساخن. Add a tea bag, ع فلت
ر الشاي في الكوب. or loose leaves, أو أوراق الشاي الطازجة. steep for a few minutes, تركه يغليج ل
بضع دقائق. then remove the bag, ثم اخرجه. or strain the leaves, أو مرر الماء عبر الأوراق. and ad
 sugar or milk, أضف السكر أو الحليب. to taste."
}
{
    "language_type": "mixed",
    "prompt": "Translate the idiom 'Break a leg' to Arabic literally (word-for-word), do not giv
 the meaning.",
    "model_name": "Qwen/Qwen2.5-7B-Instruct",
    "category": "Accuracy (Idioms)",
    "model_response": "التمزق أرجلًا"
}
{
    "language_type": "mixed",
    "prompt": "Analyze the English sentence 'The boy eats' using Arabic grammar terminology (Mub
ada and Khabar).",
    "model_name": "Qwen/Qwen2.5-7B-Instruct",
    "category": "Language Mixing (Grammar)",
    "model_response": "In Arabic grammar, the concept of Mubtada (المبتدأ) and Khabar (الخبر) is
used to identify the subject and the predicate of a sentence, respectively. However, the sentenc
 \"The boy eats\" is in English, and we need to translate it into Arabic to apply these terms ac
urately.\n\nThe English sentence \"The boy eats\" can be translated into Arabic as \"الولد يأكل\
 (al-walad ya'kul).\n\n- **Mubtada (المبتدأ)**: This is the subject of the sentence, which is th
 noun that introduces the sentence. In the Arabic translation, \"الولد\" (al-walad) means \"the
oy.\" So, \"الولد\" (al-walad) is the Mubtada.\n\n- **Khabar (الخبر)**: This is the predicate of
the sentence, which provides more information about the subject. In the Arabic translation, \"أك
ل\" (ya'kul) means \"eats.\" So, \"يأكل\" (ya'kul) is the Khabar.\n\nTo summarize:\n- **Mubtada
الولد :**(المبتدأ) (al-walad) - the boy\n- **Khabar (يأكل :**(الخبر) (ya'kul) - eats"
}
```

## Comparison Table

| Prompt Type | Fluency | Accuracy | Language Mixing | Notes |
|---|---|---|---|---|
| **English**<br><br>(*Granite*-3.0) | **Failed Constraint**<br><br>Fluent text, but failed to exclude the letter 'e'. | **High / Logical**<br><br>Correctly solved the "feathers vs. lead" trick question. | **Excellent**<br><br>Followed complex instructions (French word + English explanation) perfectly. | Strong logic and instruction following, but blind to character-level rules. |
| **Arabic**<br><br>(*ALLaM*-7B) | **Rigid / Formal**<br><br>Refused/Failed to write in Egyptian slang. | **Hallucination**<br><br>Invented fake science to justify the "born before father" logic trap. | **Failed**<br><br>Could not generate "Arabizi" | Excellent for formal Arabic only; lacks common sense reasoning. |
| **Mixed**<br><br>(*Qwen*2.5) | **Good Flow**<br><br>Sentences read naturally, though it preferred translating over mixing. | **High Reasoning**<br><br>Correctly applied Arabic grammar rules to English sentences. | **Partial**<br><br>Failed strict "every 3 words" switching, but handled semantic switching well. | Best for complex cross-lingual tasks and grammar analysis. |

## Reflection Questions Answer

**1. Which language did the model perform best on? Why?**

The **English model** performed best. It correctly solved the logic trick ("feathers vs. lead"), whereas the Arabic model failed and hallucinated fake science to explain an impossible scenario.

**2. Did Arabic output feel natural or translated?**

It felt **natural but too formal**. The model wrote good Standard Arabic (MSA) but failed to write in "Egyptian Dialect" when asked, making it sound robotic and rigid.

**3. How did the model handle mixed-language prompts?**

It handled **meaning well but failed strict rules**. It could not switch languages "every 3 words" as requested, but it successfully analyzed English grammar using Arabic terminology.

**4. Would this model be suitable for an Arabic chatbot?**

**No.** It hallucinates answers to simple logic questions and cannot understand or generate slang (dialects), which is essential for a conversational chatbot.

**5. What model would you choose for Arabic + English social media analysis?**

I would choose the **Mixed Model (Qwen)**. It demonstrated the best ability to understand and reason across both languages simultaneously.